

# Question Paper

Exam Date & Time: 04-Jan-2026 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
EIGHTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Network Security [NPT\_4148H]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) A cryptanalyst intercepts the following ciphertext, believed to be encrypted using a monoalphabetic cipher: (5)
- A) **UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUZUETU**
- She notices that the most frequent letter in the ciphertext is 'Z', which likely corresponds to 'E' in English.
- a) Analyze how frequency analysis helps the analyst to start decrypting the ciphertext.
  - b) Outline how this method becomes ineffective if the text was encrypted using a polyalphabetic cipher.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) A corporate network uses Authentication Protocol 4.0 (AP4.0), which employs symmetric key encryption for identity verification between users and servers. (5)
- a) Assess the main steps of AP4.0 and explain how it achieves authentication.
  - b) Determine one weakness of AP4.0 that could allow an attacker to impersonate a client or server.
  - c) Show one modification or variant that improves its security.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) An organization wants to send confidential documents to its international branches using PGP. (5)
- A) a) Outline how PGP combines symmetric and asymmetric encryption to ensure both confidentiality and authentication.
- b) Analyze how the web of trust model in PGP helps verify sender identity without a centralized authority.
- c) Infer one advantage and one limitation of using PGP for large-scale corporate communication.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) A company sets up a VPN between its headquarters and branch office using IPSec with Encapsulating Security Payload (ESP). (5)
- a) Apply how ESP provides both confidentiality and authentication in IP packets.
  - b) Demonstrate the ESP packet format, clearly labeling its key fields and their functions.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) After facing frequent deauthentication and disassociation attacks, a company upgrades its access points to support IEEE 802.11w. (5)
- A) a) Analyze how management frames are used in 802.11 networks and why they are vulnerable.
- b) Outline how IEEE 802.11w protects against deauthentication and disassociation attacks.
- c) Discuss the vulnerability in 802.11w networks that attackers could still exploit.

- B) With the rollout of 5G, a telecom company wants to secure its virtualized and edge computing infrastructure. (5)
- a) Demonstrate the key security features in 5G that enhance protection over 4G/EPS.
  - b) Determine the need for protecting edge computing and virtualized infrastructure in 5G environments.
  - c) Suggest one method for securing Customer Premises Equipment (CPE) or small cell devices in 5G networks.

- 4) A network architect must choose between packet filters, stateful firewalls, and an application gateway (proxy) for different services including VoIP and DNS. (5)
- A)
- a) Outline traditional (stateless) packet filtering with stateful packet inspection mention one scenario where each is appropriate.
  - b) Analyze what an application gateway (proxy) does, give one clear example, and describe one security benefit it provides over packet filters.
  - c) For DNS traffic, infer whether to allow direct outbound DNS from all hosts, force DNS through a secure resolver/proxy, or use DNS firewalling give a one-line justification.

- B) A journalist working in a restrictive country uses TOR (The Onion Router) for anonymous browsing and VPN/TLS for secure communication, but recently faced severe slowdown due to DDoS attacks on exit nodes. (5)
- a) Demonstrate how Onion Routing establishes a circuit and keys between nodes to maintain anonymity in TOR.
  - b) Determine the limitations of TLS and VPN compared to TOR when it comes to privacy and anonymity.
  - c) Examine one method used for DDoS detection in networks supporting TOR or VPN services.

- 5) A startup plans to launch its own cryptocurrency similar to Bitcoin. The development team wants to understand how block generation and mining work in Bitcoin before designing their version. (5)
- A)
- a) Outline the components of a Bitcoin block, highlighting the role of the block header and Merkle root field with an example.
  - b) Breakdown how mining is performed to find a valid block, including the concepts of target threshold and nonce adjustment.
  - c) After finding a valid block, what actions are taken by the miner and the network?

- B) A mid-sized IT company is planning to migrate its on-premises infrastructure to the cloud to reduce costs and increase scalability. The management wants to choose the right deployment model and service type. (5)
- a) Classify Public, Private, and Hybrid Cloud models in terms of ownership, scalability, and security. Give one use case for each.
  - b) Categorize IaaS, PaaS, and SaaS with examples showing how each benefits this company.
  - c) Select which cloud model and service type combination would you recommend for a company handling sensitive client data? Justify briefly.

-----End-----

| Event Name                    | Subject           |
|-------------------------------|-------------------|
| 058-MIT BLR-MAKEUP EXAM-2025- | NPT_4148H:Network |

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 10        |
| 2            | 2              | CO 2        | 10        |
| 3            | 3              | CO 3        | 15        |
| 4            | 4              | CO 4        | 15        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name                        | Subject                    |
|-----------------------------------|----------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025-2026 | NPT_4148H:Network Security |

| #     | Level       | No of Questions | Marks (%) |
|-------|-------------|-----------------|-----------|
| 1     | Remember    | 0               | 0(0%)     |
| 2     | Understand  | 0               | 0(0%)     |
| 3     | Application | 4               | 20(40%)   |
| 4     | Analysis    | 6               | 30(60%)   |
| 5     | Create      | 0               | 0(0%)     |
| 6     | Evaluate    | 0               | 0(0%)     |
| Total |             | 10              | 50 (100%) |